

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November 2013

CL203 Fluid Mechanics

Date: 26.11.2013, Tuesday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) The diameters of a pipe at the section 1-1 and 2-2 are 200 mm and 300 mm [03]
respectively. If the velocity of water flowing through the pipe at section 1-1 is 4 m/s,
find (i) discharge through the pipe, and (ii) Velocity of water at section 2-2.
- (b) Enlist different type of fluid flow and define any two. [04]
- Q - 2 (a) Define Weight density and Fluid. [02]
- (b) Calculate the specific weight, mass density and specific gravity of a liquid having a [03]
volume of 6 m^3 and weight of 44 kN.
- (c) A wooden block of specific gravity 0.75 floats in water. If the size of the block is 1 m [04]
* 0.5 m * 0.4 m, find its Metacentric height.
- (d) State and Prove Pascal's Law with neat sketch. [05]

OR

- Q - 2 (a) A U-tube manometer is used to measure the pressure of oil of specific gravity 0.85 [03]
flowing in a pipe line. Its left end is connected to the pipe and the right – limb is open
to atmosphere. The centre of the pipe is 100 mm below the level of mercury (Specific
gravity 13.6) in the right limb. If the difference of mercury in the two limbs is 160
mm, determine the pressure of the oil in the pipe.
- (b) A rectangular plate 3 m long and 1 m wide is immersed vertically in water such a way [03]
that its 3 m side is parallel to the water surface and is 1 m below it. Find (i) Total
pressure on the plate, and (ii) Position of centre of pressure.
- (c) Define Atmospheric pressure, Gauge pressure and Vacuum pressure. [03]
- (d) Enlist Newton's Law of Viscosity. [05]
- The space between two square flat parallel plates is filled with oil. Each side of the
plate is 720 mm. The thickness of the oil film is 15 mm. The upper plate, which
moves at 3 m/s requires a force of 120 N to maintain the speed. Determine (i) The

dynamic viscosity of the oil (ii) The kinematic viscosity of oil if the specific gravity of oil is 0.95.

- Q - 3 (a) Give the classification of Mouthpiece. [02]
- (b) A convergent-divergent mouthpiece having throat diameter 40 mm is discharging water under a constant head of 4 m. Determine the maximum outlet diameter to avoid separation of the flow, if the maximum vacuum pressure is 8.5 m of water. Find the discharge also. [03]
- (c) Derive an expression for discharge through a large rectangular orifice. [04]
- (d) For the following stream function calculate velocity at a point (1, 2): [05]
- (i) $\psi = 3xy$ (ii) $\psi = 3x^2y - y^3$.

OR

- Q - 3 (a) A rectangular notch 2 m wide has a constant head of 500 mm. Find the discharge over the notch, if co-efficient of discharge for the notch is 0.62. [02]
- (b) Give the classification of Weirs. [02]
- (c) A vertical sharp-edged orifice 120 mm in diameter is discharging water at the rate of 98.2 lit/sec. under a constant head of 10 m. A point on the jet, measured from the vena contracta of the jet has co-ordinates 4.5 m horizontal and 0.54 m vertical. Find the following for the orifice. [05]
- (i) Co-efficient of velocity (ii) Co-efficient of discharge, and (iii) Co-efficient of contraction.
- (d) Derive continuity equation for 3-D flow in Cartesian co-ordinates. [05]

SECTION – II

- Q - 4 (a) Enlist assumptions made in Bernoulli's equation. [02]
- (b) Water flows in a circular pipe. At section 1-1 the diameter is 0.3 m, the static pressure is 260 kPa, the velocity is 3 m/s and the elevation is 10 m above ground level. The elevation at a section 2-2 is 0 m, and the pipe diameter is 0.15 m. Find out the pressure at the section 2-2. Neglect the friction effect. Assume density of water to be 999 kg/m^3 . [05]
- Q - 5 (a) Enlist different minor losses. [02]
- (b) Define Hydraulic gradient line, Total energy line and Compound pipe. [03]
- (c) Determine the dimensions of the following quantities. [04]
- (i) Discharge (ii) Kinematic viscosity (iii) Force (iv) Specific weight
- (d) Find an expression for the drag force on smooth sphere of diameter D , moving with a uniform velocity V in a fluid density ρ and dynamic viscosity μ . [05]

OR

Q - 5 (a) An oil of specific gravity 0.92 and viscosity 0.03 poise is to be transported at the rate of 2500 lits/sec through a 1.2 m diameter pipe. Tests were conducted on a 12 cm diameter pipe using water at 20° C. If the viscosity of water at 20° C is 0.01 poise, find (i) Velocity of flow in the model, and (ii) Rate of flow in the model. [04]

(b) Enlist dimensionless numbers and define any three. [05]

(c) In a pipe of diameter 350 mm and length 75 m water is flowing at a velocity of 2.8 m/s. Find the head loss due to friction using (i) Darcy-Weisbach formula and (ii) Chezy's formula for which $C = 55$. Assume kinematic viscosity of water as 0.012 stoke. [05]

Q - 6 (a) A projectile is travelling in air having pressure and temperature as 88.3 kN/m² and -2°C. If the Mach angle is 40°, find the velocity of the projectile. Take $\gamma = 1.4$ and $R = 287$ J/kg K. [02]

(b) An aeroplane is flying at 1000 km/h through still air having a pressure of 78.5 kN/m² and temperature -8° C. Calculate on the stagnation point on the nose of the plane: [04]

(i) Stagnation pressure (ii) Stagnation temperature (iii) Stagnation density.

Take for air: $R = 287$ J/kg K and $\gamma = 1.4$

OR

(b) Derive Bernoulli's equation for compressible flow undergoing isothermal process. [04]

(c) Attempt any TWO: [08]

(i) Oil of absolute viscosity 1.5 poise and density 848.3 kg/m³ flows through a 30 cm diameter pipe. If the head loss in 3000 m length of pipe is 20 m, assuming a laminar flow, determine (i) the velocity, (ii) Reynolds number and (iii) friction factor

(ii) An oil of viscosity 9 poise and specific gravity 0.9 is flowing through a horizontal pipe of 60 mm diameter. If the pressure drop in 100 m length of the pipe is 1800 kN/m², determine (i) The rate of flow of oil (ii) Maximum velocity.

(iii) Explain displacement thickness (δ^*).
